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SYMBIOSIS CENTRE FOR INFORMATION TECHNOLOGY (SCIT), PUNE

(A constituent of the Symbiosis International University, Pune)

ESG REPORTING DASHBOARD

Plant-level emissions, renewable energy, water, waste and safety analytics with BRSR-style indicator mapping, a plant ESG scorecard and an emissions-reduction scenario (Task T20 — Mini Project)

SEMESTER-I

AI Powered Tools

No-code Apps & Dashboards

Submitted By:

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Build Evidence

The dashboard is a single self-contained HTML file (React, Recharts and SheetJS, loaded from public CDNs, so an internet connection is needed). The user uploads the Plant ESG Quarterly workbook (.xlsx); the page then shows a “Loaded 40 records” banner, record and period counters, plant and period filters, ten KPI cards, trend and comparison charts, a BRSR framework alignment panel and a plant summary matrix. Before an upload the file shows random placeholder data (Plant Alpha–Epsilon), so all screenshots below were taken after loading the real workbook.

Item	Detail
Build Link / HTML File:	Live Dashboard Link

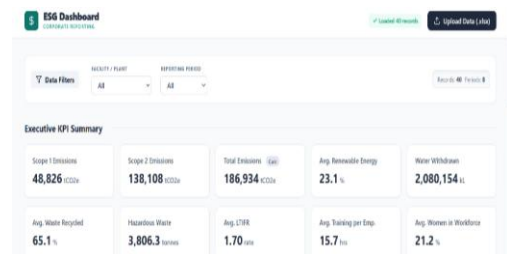


Figure 1 — Overview: upload banner, plant and reporting-period filters, record counters and ten Executive KPI cards (all plants, all periods)

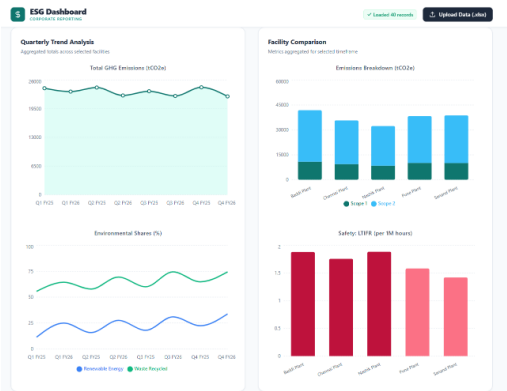


Figure 2 — Quarterly Trend Analysis and Facility Comparison (emissions breakdown by scope, LTFR by plant)

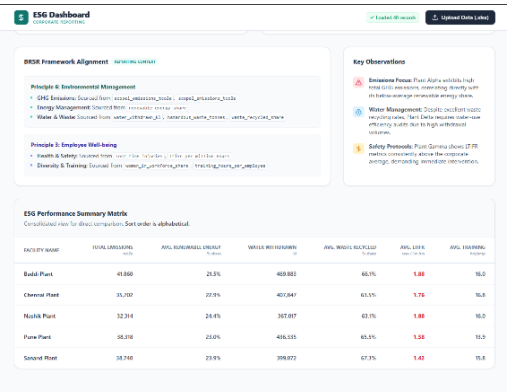


Figure 3 — BRSR Framework Alignment, Key Observations and ESG Performance Summary Matrix (the Key Observations text is static and is not used in this report; see Section 11)

1. Introduction

This report documents mini project **T20 — ESG Reporting Dashboard** for the course AI Powered Tools. The brief asks for a dashboard of emissions, renewable share, water, waste and safety by plant and quarter; a BRSR-style indicator mapping checked against the SEBI BRSR format; identification of the best and worst plants; and a draft ESG highlights page. Using the provided dataset of 5 plants × 8 quarters (40 rows), a browser-based dashboard was built with AI assistance (Gemini Canvas), checked in Excel, and then recomputed independently.

1.1 Objectives

- Build a dashboard that calculates every figure from the uploaded workbook, with a transparent indicator mapping, plant scorecard and emissions-reduction scenario (an estimate, not a guarantee).
- Verify the numbers, document AI prompts and iterations honestly, and give specific, data-backed recommendations.

Deliverables and submission. Dashboard: Build Evidence and Section 4. Indicator mapping: workbook sheet “BRSR Indicator Mapping” and Table 4. ESG highlights page: Section 5.2. AI-use log: workbook sheet “AI Use Log” (17 entries; minimum 15) and Section 3. Also submitted: the workbook with outputs added as new sheets, the build link/file, and this report.

2. Dataset Overview and Data Quality

The data is synthetic and was created for teaching; it is treated as real business data. No gold-label accuracy score was measured.

Item	Value
Size	40 rows × 13 columns (Plant ESG Quarterly sheet); all 13 expected columns present
Plants	Pune, Nashik, Chennai, Baddi, Sanand (8 quarters each, Q1 FY25 to Q4 FY26)
Integrity checks	0 blank cells; 0 duplicate plant–quarter keys; “Loaded 40 records, 8 periods” shown by the dashboard (Figure 1)
Name issue found	An extra space in the Nashik plant name broke an Excel formula (log 10); corrected before analysis
Plausibility flags	10 of 40 rows show 0 lost-time injuries with LTIFR > 0; headcount swings by 19% per quarter on average within a plant (max 43%)

Table 1 — Dataset profile and data-quality result

3. AI-Assisted Development Process

Table 2 shows four build rounds. Gemini Canvas produced the dashboard, the mapping and the draft text; Microsoft Excel was used for verification (log 9–11, 16). The AI Use Log holds 17 entries, including one rejected output (entry 2).

Round (log #)	Problem found	Prompt / process change and reason	Outcome
1 · Initial build (1–2)	First structure was fine, but after the Excel upload several charts were blank	Reviewed against the brief; compared the workbook’s real column names with the fields the dashboard expected	Partly kept (entry 2 rejected)
2 · Data mapping (3–4)	Dashboard fields did not match the workbook headers	Gave the exact column names of the Plant ESG Quarterly sheet and told the tool not to invent unavailable fields	Kept
3 · Reported fields only (5–8)	Tool tried to calculate renewable share, recycled share and LTIFR from columns that do not exist	“Use X directly ... do not calculate” for each field; retested that KPI cards, filters and charts populate	Kept as final build
4 · Mapping and highlights (12–15, 17)	Mapping and highlights had to stay inside the dataset and be balanced	Restricted to indicators present; asked for improvements and attention areas; linked each recommendation to a metric; final checklist review	Kept

Table 2 — Documented prompt-improvement rounds (source: AI Use Log)

The Excel checks (log 9–11, 16) were the independent layer: SUMIFS/AVERAGEIFS plant and FY tables built from the source sheet. AI output was treated as something to verify; the dashboard was accepted only after its cards matched those tables (Section 11).

Element	Origin	Status
KPI definitions (sums for quantities; simple averages for shares and rates; reported LTIFR)	Specified in my prompts (log 5–7); checked in Excel (log 9, 11)	Adopted; no derived fields
BRSR-style mapping (12 metrics, 8 topics)	Drafted by Gemini (log 12); kept at topic level	Adopted; official format check still to be noted
Scorecard weights 25/15/15/10/10/25 and 35% renewable target	Proposed by Claude while drafting this report; not in the brief or dashboard	Management assumptions: student to confirm and log (row 18)
“Key Observations” panel in the dashboard	Generated by Gemini as static text	Not used; flagged in Section 11
Highlights and recommendations	First drafts by Gemini (log 14–15); numbers re-derived and targets set in this report	Student to review before submission

Table 3 — Data-derived findings, AI suggestions and human decisions

4. Application and Dashboard Methodology

4.1 Dashboard components

Header with Excel upload and a “Loaded 40 records” banner; plant and reporting-period filters with record and period counters; Executive KPI Summary (ten cards); Quarterly Trend Analysis (total GHG emissions; renewable and recycled shares); Facility Comparison (Scope 1/Scope 2 breakdown; LTIFR by plant); BRSR Framework Alignment panel; Key Observations panel; and an ESG Performance Summary Matrix by plant. The scorecard (Section 8) and scenario (Section 9) are report-stage analyses, not dashboard tabs.

4.2 KPI definitions and BRSR-style mapping

Metric	Source column	Aggregation used	BRSR-style topic
Scope 1 / Scope 2 emissions	scope1_ / scope2_ emissions_tco2e	Sum	GHG emissions
Total emissions	Scope 1 + Scope 2	Sum of both scopes	GHG emissions
Renewable energy share	renewable_energy_share	Simple average × 100 (reported, not recomputed)	Energy
Water withdrawn	water_withdrawn_kl	Sum (kL)	Water
Waste recycled share	waste_recycled_share	Simple average × 100	Waste
Hazardous waste	hazardous_waste_tonnes	Sum (tonnes)	Waste
Lost-time injuries *	lost_time_injuries	Sum (count)	Occupational health & safety
LTIFR	ltifr_per_million_hours	Simple average (reported, not recomputed)	Occupational health & safety
Training hours per employee	training_hours_per_employee	Simple average	Employee development
Women in workforce	women_in_workforce_share	Simple average × 100	Diversity & inclusion
Headcount *	headcount	Average	Workforce

Table 4 — KPI definitions and topic mapping (* in the workbook analysis; not a dashboard card). Mapping is BRSR-style at topic level, not a claim of exact BRSR disclosure fields

4.3 Scorecard and emissions scenario

Scorecard. Each plant receives six 0–100 sub-scores by min–max normalisation across the five plants (higher is better for renewable and recycled share; lower is better for emissions, water, hazardous waste and LTIFR, so these are inverted), using FY25–FY26 combined values. Weights: emissions 25%, LTIFR 25%, renewable 15%, water 15%, recycled 10%, hazardous waste 10%. Grades are relative (A ≥ 80, B ≥ 65, C ≥ 50, else D). **Scenario.** For each plant below a 35% renewable share, Scope 2 emissions are assumed to scale with the non-renewable share: $\text{reduction} = \text{FY26 Scope 2} \times (1 - (1 - \text{new share}) \div (1 - \text{current share}))$, at 25%, 50% or 100% closure of the gap to 35%. Weights, grades and the 35% target are management assumptions, not dataset facts.

4.4 Error handling and transparency

In the HTML file reviewed, headers and plant names are trimmed and matched case-insensitively; an error message is shown if the sheet is empty or the plant/quarter columns are missing; and non-numeric cells become 0 rather than being reported, so a blank cell would be silently counted as zero. No blank cells exist in this workbook, but there is no missing-value count or validation panel. Placeholder data appears until a workbook is uploaded.

5. Overall Analytical Results

5.1 Network totals (all 40 plant-quarters)

Metric	Value	Basis
Total emissions	186,934 tCO ₂ e	Scope 1 48,826 + Scope 2 138,108 (FY25 96,317; FY26 90,617)
Scope 2 share of emissions	73.9%	138,108 ÷ 186,934
Average renewable share	23.1%	FY25 16.9% → FY26 29.35%
Water withdrawn	2,080,154 kL	FY25 1,078,030; FY26 1,002,124
Average waste recycled share	65.1%	FY25 59.6% → FY26 70.7%
Hazardous waste	3,806.3 t	FY25 1,878.2; FY26 1,928.1
Lost-time injuries / average LTIFR	49 / 1.70	FY25 21 / 1.77; FY26 28 / 1.64
Training / women in workforce (avg)	15.7 h / 21.2%	Simple averages of the two fiscal-year averages

Table 5 — Overall verified KPIs (the dashboard cards show the same values, Figure 1)

5.2 Draft ESG highlights — FY26 vs FY25

Indicator (all plants)	FY25	FY26	Change	Reading
Total emissions (tCO ₂ e)	96,317	90,617	−5,700 (−5.9%)	Improved
Renewable energy share (avg)	16.9%	29.4%	+12.5 pp	Improved
Water withdrawn (kL)	1,078,030	1,002,124	−75,906 (−7.0%)	Improved
Waste recycled share (avg)	59.6%	70.7%	+11.1 pp	Improved
Women in workforce (avg)	18.4%	23.9%	+5.5 pp	Improved
Average LTIFR	1.77	1.64	−0.13 (−7.4%)	Improved
Hazardous waste (t)	1,878.2	1,928.1	+49.9 (+2.7%)	Needs attention
Lost-time injuries (count)	21	28	+7 (+33%)	Needs attention
Training hours per employee (avg)	16.8	14.6	−2.25 (−13.4%)	Needs attention

Table 6 — ESG highlights (quantities are totals; shares and rates are simple averages)

- **Environment:** emissions fell 5.9% (Scope 2 −7.0%, Scope 1 −2.9%), a smaller gain than the 12.5-point rise in renewable share suggests. Water and recycling improved; hazardous waste rose 2.7%.
- **Social:** women's share and LTIFR improved, but injuries rose 33% and training fell 13.4%.

6. Plant Performance

Plant	Emissions (tCO ₂ e)	Renew. share	Water (kL)	Recycled share	Haz. waste (t)	LTIs	Avg LTIFR
Pune	38,318	23.0%	436,335	65.5%	796.5	9	1.58
Nashik	32,314	24.4%	367,017	63.1%	665.8	10	1.88
Chennai	35,702	22.9%	407,847	63.5%	767.5	11	1.76
Baddi	41,860	21.5%	469,883	66.1%	811.1	5	1.88
Sanand	38,740	23.9%	399,072	67.2%	765.4	14	1.42

Table 7 — Plant scorecard values, FY25–FY26 combined (green = best, red = worst on that metric)

Nashik is the strongest environmental performer (lowest emissions, water and hazardous waste; highest renewable share) and **Baddi the weakest** (worst on four of five environmental measures). Safety is mixed: Sanand has the lowest LTIFR (1.42) yet the most injuries (14); Baddi has the fewest injuries (5) but an LTIFR of 1.88, level with Nashik (1.884 vs 1.880). Because of the LTIFR inconsistency in Section 2, safety rankings are provisional.

7. Plant-Level Insights

Table 8 shows the sharpest FY25 → FY26 plant movements. Each is taken from the plant-by-year recomputation of the source sheet.

Plant · Indicator	FY25	FY26	Observation
Nashik · Renewable share	17.0%	31.8%	Largest renewable gain (+14.8 pp); emissions −6.6%
Pune · Emissions	20,107	18,211	Largest cut (−9.4%), but training fell 17.25 → 10.5 h (−39%)
Sanand · LTIs	4	10	Six of the group's seven extra injuries, while its LTIFR fell 1.51 → 1.33
Baddi · Hazardous waste (t)	379.2	431.9	Largest increase (+52.7 t, +13.9%); only Sanand cut it (−80.0 t)
Chennai · LTIFR	1.53	1.98	Only plant with a higher LTIFR (+29%); also the only one with higher water use (+3.1%)

Table 8 — Selected plant × indicator comparisons (FY25 vs FY26)

Sanand's emissions fell only 2.5% despite a 13.7-point renewable gain, the weakest response of the five plants. Across all 40 plant-quarters, renewable share and Scope 2 emissions are negatively correlated ($r = -0.38$), an indicative rather than strong link.

8. Plant ESG Scorecard

Management assumptions – configurable: emissions 25%, LTIFR 25%, renewable 15%, water 15%, recycled 10%, hazardous waste 10%. These weights were proposed by AI while drafting this report; they are not derived statistically from the data.

Plant	Emissions	Renewable	Water	Recycled	Haz. waste	LTIFR	Weighted total (grade)
Sanand	33	83	69	100	32	100	69.0 (B)
Nashik	100	100	100	0	100	0	65.0 (B)
Pune	37	52	33	58	10	65	45.0 (D)
Chennai	65	48	60	9	30	28	43.1 (D)
Baddi	0	0	0	73	0	1	7.5 (D)

Table 9 — Plant scorecard (relative 0–100 sub-scores, FY25–FY26 combined)

Sanand ranks first on safety and recycling, Nashik on every environmental measure except recycling, and Baddi is last. The top two are close: on FY26 values alone Nashik leads (75.2 vs 72.6), so the order of the top two depends on the weights and period. Min–max scores stretch small gaps: Nashik and Baddi differ by 0.004 in LTIFR yet score 0 and 1 on it. The scorecard is an ordering under these weights, not an objective “best plant”, and should be read with Table 7.

9. Emissions-Reduction Scenario

Closing **50% of each plant's gap to a 35% renewable share** gives an estimated/potential reduction of **2,689 tCO₂e** a year (3.0% of FY26 emissions of 90,617 t). This is a scenario estimate, not a guaranteed saving: 25% closure gives 1,345 t and 100% closure gives 5,378 t (5.9%). At 100% closure the plants contribute: Baddi 1,687 t, Pune 1,208 t, Chennai 1,064 t, Sanand 869 t and Nashik 550 t; Baddi and Pune together supply 54%.

- All five plants are below 35% (FY26 average: Nashik 31.8%, Sanand 30.8%, Chennai 29.0%, Pune 28.5%, Baddi 26.8%); the best plant-quarter reached 37% (Sanand, Q4 FY26), so 35% is achievable on current evidence.
- The estimate assumes Scope 2 moves in proportion to the non-renewable share, that renewable share refers to electricity, and that consumption and grid emission factors stay unchanged. It ignores Scope 1, cost and supply constraints, and the data link is weak ($r = -0.38$).

10. Business Recommendation

The recommendations below are for the Head of Sustainability / Plant Operations. Targets are proposals derived from the verified FY26 figures; savings are estimated/potential, not guaranteed.

Sr No.	Action	Evidence (from the data)	Metric and proposed target (FY27)
1	Sanand safety review: audit the Q2–Q4 FY26 incidents and confirm controls	LTIs 4 → 10 at Sanand, six of the group's seven extra injuries; group LTIs 21 → 28	Sanand ≤ 4 LTIs; group ≤ 21
2	Baddi decarbonisation: prioritise renewable electricity procurement	Baddi FY26 emissions 20,374 t (highest); renewable 26.8% vs Nashik 31.8%; Scope 2 14,981 t; scenario –1,687 t at 35%	Renewable ≥ 35%; emissions ≈ 18,687 t (–1,687 t, –8.3%)
3	Restore training: reinstate safety-linked training, starting at Pune	Group 16.8 → 14.6 h while injuries rose 33%; Pune 17.25 → 10.5 h (–39%)	≥ 16.8 h per employee at every plant
4	Baddi water efficiency: adopt Nashik's practices	Baddi FY26 222,090 kL vs Nashik 171,379 kL (best)	Baddi ≤ 200,425 kL (–9.8%)
5	Hazardous waste review: root-cause at Baddi, Pune and Chennai	Baddi +52.7 t, Pune +32.7 t, Chennai +27.5 t; Sanand –80.0 t	Group ≤ 1,878 t (FY25 level)
6	Data governance: reconcile LTIFR with hours and LTIs; stabilise headcount; fix quarter sorting; remove static observations	10 of 40 rows with 0 LTIs but LTIFR > 0; headcount volatility 19%; Figures 2–3	0 unreconciled rows; dashboard re-tested before next cycle

Table 10 — Prioritised recommendations linked to measurable indicators

Overall, the environmental trend is positive but modest in carbon terms (–5.9% emissions against +12.5 pp renewable share), while the social indicators moved the wrong way (injuries +33%, training –13.4%). Safety and training come first, followed by a targeted renewable push at Baddi and Pune, where the scenario shows the largest reductions.

11. Verification and Correctness

Check	What was checked	Result
Data integrity	40 rows × 13 columns; 0 blanks; 0 duplicate plant–quarter keys	Pass
Dashboard KPI cards (Figure 1)	10 of 10 cards match an independent recomputation: Scope 1 48,826; Scope 2 138,108; total 186,934; renewable 23.1%; water 2,080,154 kL; recycled 65.1%; hazardous 3,806.3 t; LTIFR 1.70; training 15.7 h; women 21.2%	Pass (values read from screenshot)
Summary matrix (Figure 3)	Emissions, renewable share, water, recycled share and LTIFR for all five plants match Table 7	Pass
Excel plant and FY25/FY26 tables	SUMIFS/AVERAGEIFS results match the recomputation (e.g. Baddi 41,860 t; LTIs 21 → 28)	Pass
Scorecard and scenario	Computed from the source sheet in Python for this report; not yet in the workbook	Add as a workbook sheet
Blank charts after upload	Field names mismatched; fixed with exact column names (log 2–8)	Fixed
Nashik formula error	Extra space in the plant name; corrected and table rechecked (log 10, 16)	Fixed
Quarter order in charts	Figure 2 axis reads Q1 FY25, Q1 FY26, Q2 FY25 ...: alphabetical sort, not time order	Open: sort by FY, then quarter
Key Observations panel	Figure 3 text names Plant Alpha, Delta and Gamma, which are not in the dataset; it is static text, not generated from the upload	Open: remove or regenerate
LTIFR vs LTI counts; headcount	10 of 40 rows with 0 LTIs but LTIFR > 0; implied hours reach about 3,170 per employee per quarter; headcount swings 19% a quarter	Flagged as data caveat

Table 11 — Verification evidence, errors found and their status

Verification covered every plant and FY aggregate but not every filter combination in the dashboard, and with no Gold Labels sheet no gold-label accuracy was measured. KPI-card values were read from the screenshot in Figure 1, not from a fresh run of the file.

12. Responsible AI and Limitations

- **Errors found and corrected:** blank charts from mismatched fields; AI attempts to derive renewable share, recycled share and LTIFR from missing columns; a plant-name spacing error. **Still open:** quarter order and the static Key Observations text.
- **AI limitations:** the tool assumed column names, tried to derive unavailable fields, and wrote static “observations” about plants that do not exist in the data. Every output needed inspection.
- **Free-tools rule:** the brief says free tools only; the log records Gemini (paid), Gemini 3.1 Pro. This is disclosed here and should be confirmed with the faculty. Only synthetic data was used and no personal or confidential data was entered into any tool.
- **Management assumptions and data limits:** scorecard weights, grades and the 35% target are configurable choices; 40 synthetic rows, simple (unweighted) averages, no production volumes, so no intensity metrics; the BRSR mapping is topic-level and still needs a line-by-line check against the official SEBI format.

13. Personal Reflection

The tool produced a polished dashboard quickly, but it was blank on real data because it assumed column names; only checking the workbook exposed this. Prompts that named exact fields and forbade calculation worked far better than broad requests, and the rejected output taught me as much as the accepted ones. Excel checks caught a small naming error, and the final review showed that numbers can reconcile perfectly while the data itself (LTIFR against injuries) still hides quality problems. Next time I would verify earlier, log evidence as I go, and test sorting and edge cases before trusting a chart